# Tiancong (Lindsay) Li

Research Interests: LLM Agents · Tool Learning · Efficient and On-Device LLMs

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## Education

**University of California, Irvine** **Sep. 2025 – Dec. 2026**

*M.S. in Embedded and Cyber-Physical Systems | GPA: 3.78/4.00* *Irvine, CA*

- **Coursework:** AI Accelerator Design (in progress), Embedded Systems Modeling and Design, Real-Time Systems, Internet of Things Systems and Software, Wireless Sensor and Actuator Networks

**University of California, Irvine, Division of Continuing Education** **Sep. 2024 – Jun. 2025**

*Certificate Program in Electrical Engineering and Computer Science | GPA: 3.93/4.00* *Irvine, CA*

- **Coursework:** Deep Learning, Computer Vision

**Dalian University of Technology** **Sep. 2021 – Jun. 2025**

*B.Eng. in Software Engineering | Average: 85/100* *Dalian, China*

- **CS Coursework:** Machine Learning, Introduction to Artificial Intelligence, Data Structures and Algorithms
- **Math Coursework:** Probability and Statistics, Linear Algebra, Calculus, Discrete Mathematics

## Publications

**Tiancong Li**, Wenzhuo Zhou. First-author paper on efficient tool-using LLM agents. Under review at ACL Rolling Review (ARR), October 2026.

## Research Experience

**Parameter-Efficient Tool Agents with Tool Tokens** **Oct. 2025 – Present**

*First Author | Advisor: Prof. Wenzhuo Zhou, Department of Statistics, UC Irvine* *Irvine, CA*

- Designed a modular tool agent that splits tool selection, tool calling and answer writing across three lightweight models, letting 0.6B models work over 47K real-world APIs.
- Outperformed ToolGen-8B by **10.9 points** of grounded success with 4.4× fewer total parameters, using **6× less training compute and 11× less inference compute**.
- Diagnosed why small tool agents fail: agent training suppresses the tools its trajectories never call (85% of the tool set), cutting top-1 tool selection from **89% to 12%** at 0.6B.
- Fixed it by design, keeping tool selection out of agent training, which lifted correct-tool rate to **94.5%**, above ToolGen-8B's 86.9%.
- Showed that answer writing needs no fine-tuning: an off-the-shelf instruction-tuned model beats every answer module fine-tuned on the benchmark, adding **12.7 points** of grounded success at 0.6B.
- Introduced a **grounded-success** metric that requires a correct tool call behind every accepted answer, revealing that 52% of a BM25 baseline's "successes" never used a correct tool; results hold across seeds, a stricter judge and a blind human evaluation.

## Selected Projects

**On-Device Robot Manipulation Policies** | *MECPS Capstone (Synaptics)* **Mar. 2026 – Present**

- Led code and model training in a 4-person team, delivering an end-to-end pipeline from robot data collection to **on-device inference on the Synaptics SL2619 NPU**.
- Benchmarked 4 robot policies (ACT, SmolVLA, Diffusion Policy, Pi0.5) and deployed the best one to the edge NPU, reaching **10/10 pick-and-place success**.
- Building a Gemma-based voice-command module for a medication-dispensing application with industry partner NousLogic.

**Intelligent Traffic Detection Drone** | *Dalian University of Technology* **Apr. 2022 – Apr. 2023**

- Built a Raspberry Pi + STM32 drone combining deep-learning image recognition with sensor fusion for real-time traffic event detection; added a Blockly visual-programming interface. **3rd Prize, National Computer Design Competition**.

## Technical Skills

**ML & LLMs**: PyTorch, Hugging Face Transformers, DeepSpeed, LoRA, constrained decoding, LLM-as-judge evaluation
**Programming**: Python, C, C++, MATLAB, Linux, Git
**Systems**: ARM Cortex-M, STM32, ESP32, Raspberry Pi, FreeRTOS, TensorFlow Lite (edge inference)